Development Across the Lifespan

How people change from infancy to old age — thinking, attachment, identity and ability — and what that means for coaching, careers and the myth of generational workers.

Isagi · November 2026 · Spine: Ciccarelli & White, *Psychology* (4th ed.), Ch. 8 — rewritten, updated and applied

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A manager typically leads people aged 22 to 58. They differ in what they know, how fast they learn new systems, what they want from work and how they handle setbacks. Some of that is personality (Part 11). Much of it is where people are in a life that follows broadly predictable stages. Developmental psychology studies those changes. This part covers the classic theories, which ones survived testing, and the management lessons: how to coach, how ability changes with age, and why "millennials want…" slides deserve suspicion.

1 The big questions and how they are studied

Two debates run through the field. **Nature versus nurture**: how much of development is driven by genes and how much by environment? The modern answer is that they interact. Genes set ranges and tendencies, and environment determines where in the range a person ends up; environments can even change which genes are switched on (**epigenetics**). And **continuity versus stages**: is development smooth and gradual, or a series of distinct steps? Most traits change gradually, though some stage theories remain useful as maps.

Three research designs, each with a trap:

- **Cross-sectional** — compare different age groups at one time. Quick, but age differences get mixed up with **cohort effects**: differences due to the era people grew up in, not their age.

- **Longitudinal** — follow the same people for years. Shows true change, but slow, expensive and prone to drop-outs.
- **Cross-sequential** — follow several age groups over time; combines both and separates age from cohort.

WORKED EXAMPLE 1 · AGE OR ERA?

A 2026 survey of a sales force finds that reps aged 25 use the CRM app 40 times a week, and reps aged 50 use it 10 times. HR concludes: "People get worse with technology as they age."

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Cross-sectional finding:  25-year-olds 40 / week
                          50-year-olds 10 / week
Possible explanations:
  Age effect:     ability to adopt tech declines with age
  Cohort effect:  50-year-olds grew up without smartphones
  Role effect:    50-year-olds are senior; juniors log more
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A cross-sectional survey can't tell these apart. To separate them, follow the same people over time, compare people of the same age in different roles, or train both groups and measure the change. The likely answer is mostly cohort and role — and the remedy for cohort is training, not hiring younger.

2 Beginnings

Prenatal development runs from the fertilised egg (the **zygote**) through the **embryo** stage, when the major organs form, to the **foetus**. Substances that can harm the developing baby are called **teratogens**; alcohol is among the most common, and can cause lifelong learning and behavioural problems. Newborns arrive with reflexes such as grasping and rooting (turning toward a touch on the cheek), and with senses ready to learn: within days they prefer their mother's voice and faces over other patterns. The brain grows rapidly in the first years, forming far more connections than it keeps, then pruning those that go unused — plasticity in action (Part 1).

3 How thinking develops

Piaget's four stages

Jean Piaget, observing children from the 1920s, argued that thinking develops through four qualitatively different stages. Children are not just adults who know less; they think differently.

Stage (approx. age)	What changes
Sensorimotor (0–2)	Learning through the senses and action. **Object permanence** — knowing a hidden object still exists — develops.
Preoperational (2–7)	Language and symbols, but thinking is **egocentric** (hard to take another's view) and fails at **conservation**: a child thinks a tall thin glass holds more than a short wide one with the same water.
Concrete operational (7–11)	Logical thinking about concrete things; conservation mastered.
Formal operational (12+)	Abstract and hypothetical reasoning: "what if" and systematic testing of ideas.

Piaget's broad sequence has held up, but later research showed he underestimated infants (some understand object permanence earlier than he thought), that progress is more gradual and uneven than neat stages suggest, and that many adults reason formally only in areas they know well.

Vygotsky: learning with others

Lev Vygotsky stressed that children learn through interaction with more skilled people. His **zone of proximal development (ZPD)** is the gap between what a learner can do alone and what they can do with help. Teaching works best inside that zone, through **scaffolding**: support that is gradually withdrawn as competence grows. This is the best single model of field coaching. A rep who can already run a routine dealer visit alone doesn't need you on that call. He needs you on the first collections conversation with a difficult dealer — then less on the second, and not at all by the fifth.

4 Attachment, temperament and parenting

In the 1950s Harry Harlow raised infant monkeys with two artificial "mothers": one made of wire that gave milk, one covered in soft cloth that gave none. The infants spent most of their time clinging to the cloth mother and ran to her when frightened. Comfort, not just food, drives the bond. Mary Ainsworth's **Strange Situation** test — briefly separating a toddler from the parent and observing the reunion — identified attachment patterns: **secure** (upset at separation, easily comforted on return), **avoidant**, **ambivalent** (clingy and hard to soothe), and later **disorganised**. Secure attachment is linked to better social adjustment later, though the links are moderate, not destiny.

Babies also differ from birth in **temperament**: some are "easy", some "difficult", some "slow to warm up". Temperament is partly inherited and partly stable over time — an early root of adult personality.

Diana Baumrind described parenting styles: **authoritarian** (strict rules, little warmth or explanation), **permissive** (warm, few demands), **authoritative** (clear expectations with warmth and explanation) and, added later, **uninvolved**. In Western studies authoritative parenting is linked with the best outcomes; the picture varies more across cultures. The parallel with management styles is obvious and reasonably well supported: high standards plus warmth and explanation usually beats either harshness or indulgence.

5 Erikson's eight stages

Erik Erikson proposed that each life stage brings a central social conflict to resolve. His theory is hard to test precisely, but it remains a useful map of the concerns people bring to work at different ages.

Age	Conflict	What's at stake
0–1	Trust vs mistrust	Whether the world is reliable
1–3	Autonomy vs shame and doubt	Doing things for oneself
3–6	Initiative vs guilt	Taking the lead in play and plans
6–12	Industry vs inferiority	Feeling competent at tasks
Teens	Identity vs role confusion	Who am I, what do I stand for
20s–30s	Intimacy vs isolation	Close relationships and commitment
40s–60s	Generativity vs stagnation	Contributing to the next generation
65+	Integrity vs despair	Making peace with the life lived

The stage most useful to managers is **generativity**. Many people in their forties and fifties want to mentor and leave something behind. A senior rep who has plateaued in sales may be your best trainer, if the role is offered as recognition rather than as a demotion.

6 Adolescence and moral reasoning

Adolescence begins with **puberty** and ends, culturally, with taking on adult roles — later today than a generation ago. The brain's reward systems become highly active in the teens while the **prefrontal cortex**, which handles planning and self-control, keeps maturing into the mid-twenties. That mismatch helps explain risk-taking, especially with peers watching. It also means that the 22-year-old trainee's brain is still finishing development.

Lawrence Kohlberg described moral reasoning in three levels: **preconventional** (avoid punishment, gain reward), **conventional** (follow rules and social expectations), and **postconventional** (reason from principles such as justice, even against rules). Critics, notably Carol Gilligan, argued it overvalued abstract justice compared with care for relationships, and later research found that how people reason about dilemmas predicts their actual behaviour only modestly. Most compliance problems in sales are better explained by incentives and norms (Parts 4 and 10) than by stages of moral reasoning.

7 Adulthood and ageing

Adulthood is usually divided into early (20s–30s), middle (40s–60s) and late adulthood. Physical capacities peak in the twenties and decline gradually. The more important news for managers is about mental ability, which does not decline as a block. Raymond Cattell and John Horn distinguished two kinds:

- **Fluid intelligence** — speed, working memory and solving new problems. It peaks in early adulthood and declines gradually from the thirties.
- **Crystallised intelligence** — accumulated knowledge, vocabulary and expertise. It keeps rising or holds steady well into the sixties.

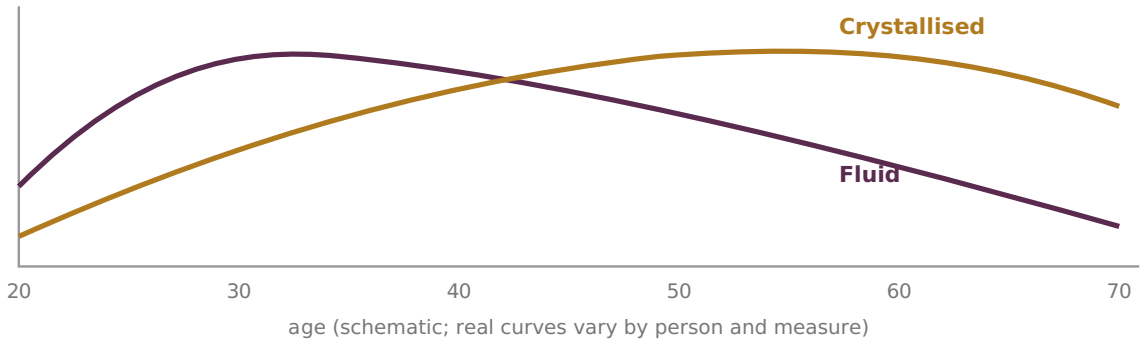


Figure 1. Two kinds of intelligence across adulthood. Fluid ability peaks early; crystallised knowledge keeps growing.

The management implication: older employees often learn a new software system more slowly, but bring judgement, relationships and pattern recognition that take decades to build. Teams that pair them with younger colleagues, each covering the other's weaker side, beat teams that are uniform in age.

The **midlife crisis** — a dramatic upheaval around 40 to 50 — is much less common than its fame suggests; most adults report midlife as stable, and large surveys find life satisfaction tends to dip in midlife but only modestly. In later life, people who stay active and socially connected tend to fare better (**activity theory**), and older adults increasingly focus on emotionally meaningful relationships over new contacts.

Death and grief

Elisabeth Kübler-Ross's 1969 description of five stages of dying — denial, anger, bargaining, depression, acceptance — became famous and is often applied to grief and even to organisational change. Research does not support a fixed sequence. People move between these feelings in different orders or skip some, and studies of bereavement find that many people are resilient, returning to normal functioning within months. Using "the five stages" to manage a restructuring is folklore. Offer time, flexibility and support, without expecting a timetable.

8 Held up?

Claim	Verdict	Why
Delaying a marshmallow at age 4 predicts adult success	**Much weaker**	A 2018 replication with about 900 children found a smaller link, which mostly disappeared after controlling for family background and early ability.
Grief follows five fixed stages	**No**	No fixed sequence; resilience is common.
Generations (millennials, Gen Z) want fundamentally different things at work	**Mostly not**	Meta-analyses find small generational differences in work attitudes; most gaps reflect age and career stage, which every generation passes through.
Fluid ability declines, crystallised knowledge grows	**Robust**	Consistent across longitudinal and cross-sectional studies.
Midlife crisis is near-universal	**No**	A minority experience it; satisfaction dips modestly on average.

Summary

- Genes and environment **interact**. Cross-sectional studies confuse age with **cohort**; longitudinal and cross-sequential designs separate them.
- **Piaget**: sensorimotor, preoperational, concrete and formal operational stages. The sequence holds, but he underestimated infants and stages are less tidy.
- **Vygotsky**: teach within the **zone of proximal development** with **scaffolding** that is withdrawn as skill grows — the model for field coaching.
- Attachment is built on comfort (Harlow); Ainsworth's patterns predict later adjustment moderately. Temperament is partly inborn. Authoritative parenting — standards plus warmth — has the best outcomes in most studies.
- **Erikson's** eight stages map life's central concerns; **generativity** in midlife makes experienced staff good mentors.
- The prefrontal cortex matures into the mid-twenties. Kohlberg's moral stages predict behaviour only modestly.
- **Fluid intelligence** declines from the thirties; **crystallised intelligence** grows into the sixties. Mixed-age teams cover both.
- The marshmallow test, five stages of grief, the universal midlife crisis and big generational differences at work don't hold up.

Practice problems

P1. A survey finds that 55-year-old managers are less willing to relocate than 28-year-olds. List one age explanation, one cohort explanation and one life-stage explanation.

P2. Describe how you would coach a new rep on handling overdue payments using the zone of proximal development and scaffolding, across four dealer visits.

P3. A 52-year-old rep has plateaued on volume but knows every dealer in the region. Which Erikson stage is relevant, and what role would you offer him?

P4. A consultant's slide says "Gen Z won't stay more than a year because they value purpose over pay." What would you ask before acting on it?

P5. You are forming a team to launch a new digital ordering app with dealers. Using fluid and crystallised intelligence, how would you staff it?

P6. After a restructuring, HR plans to "move people through the five stages of change by week six". What's wrong with this?

Glossary

Cohort effect

Differences due to the era people grew up in, not their age.

Cross-sectional / longitudinal

Compare age groups once / follow the same people over time.

Epigenetics

Environmental influences on which genes are switched on.

Teratogen

A substance that can harm prenatal development.

Object permanence

Knowing that objects exist when out of sight.

Egocentrism

Difficulty taking another person's point of view.

Conservation

Understanding that quantity stays the same when shape changes.

Zone of proximal development

The gap between what a learner can do alone and with help.

Scaffolding

Support that is gradually withdrawn as competence grows.

Attachment

The emotional bond between an infant and caregiver.

Temperament

Inborn tendencies in mood and reactivity.

Authoritative parenting

High expectations combined with warmth and explanation.

Generativity

Erikson's midlife drive to contribute to the next generation.

Fluid / crystallised intelligence

Speed and new problem-solving / accumulated knowledge.

Activity theory
The view that staying active and engaged supports healthy ageing.

Answer key

P1. Age: less appetite for upheaval with age. Cohort: grew up when careers were more local and stable. Life stage: children in school, ageing parents, a spouse's job, a home owned.

P2. Visit 1: you lead the conversation while he observes. Visit 2: he leads; you step in only if it stalls. Visit 3: he leads and you stay silent, debriefing afterwards. Visit 4: he goes alone and reports back. Each step stays just beyond what he can do alone.

P3. Generativity vs stagnation. Offer a mentoring or trainer role — onboarding new reps, joint visits in key accounts — framed and rewarded as recognition.

P4. Is the evidence comparing Gen Z with earlier generations *at the same age*? Young employees of every generation have changed jobs more often. What's our own data on early attrition by age, cohort and manager?

P5. Pair younger staff (high fluid ability, quick on the app) with senior reps (crystallised knowledge of dealers, credibility, objections). Each dealer visit should have both skill sets in the room.

P6. Research doesn't support fixed stages or timetables for adjustment; many people adapt quickly and others move back and forth. Offer information, support and flexibility, and measure how people are actually doing.

Sources: chapter structure follows Ciccarelli & White, *Psychology*, 4th ed. (Pearson), Ch. 8; text and figures are original. Marshmallow replication: Watts, Duncan & Quan, *Psychological Science*, 2018. Grief and resilience: Bonanno, *American Psychologist*, 2004. Generational differences: Costanza et al., *Journal of Business and Psychology*, 2012. Fluid and crystallised intelligence: Horn & Cattell, 1967, and later longitudinal work. Worked-example figures are illustrative.